



Recurrent Graph Neural Network Hybrid Model for Spatio-Temporal Traffic Flow Prediction in Intelligent Transportation Systems

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Abstract

Traffic congestion is a growing challenge in urban environments, driven by increasing population and vehicle density, leading to significant economic and societal impacts. Accurate traffic forecasting is a key component of Cooperative Intelligent Transportation Systems (C-ITS), enabling proactive traffic management strategies such as adaptive signal control and dynamic routing. However, existing approaches often struggle to capture spatial dependencies in road networks and long-range temporal dynamics in traffic data. This paper proposes TETRA, a hybrid spatio-temporal traffic forecasting model that integrates Graph Convolutional Networks (GCNs) with Extended Long Short-Term Memory (xLSTM). By incorporating matrix-based memory and memory mixing, xLSTM enables the model to capture complex multi-timescale temporal patterns, including congestion propagation and delayed recovery dynamics, which are not well represented by conventional recurrent models. The proposed approach is evaluated on a real-world urban traffic dataset and the widely used METR-LA benchmark. Experimental results show that TETRA outperforms or matches established baseline models representative of the main spatio-temporal paradigms, with the most pronounced gains at medium- and long-term horizons, achieving up to 13.0% lower MAE, 20.0% lower RMSE, and 6.0% higher R^2 on a real-world dataset relative to the strongest investigated baseline at each horizon. Statistical analysis confirms that these improvements are robust across prediction horizons. Additional evaluations, including ablation and sensitivity analyses, demonstrate the effectiveness and scalability of the proposed model.

Keywords Traffic flow prediction · Neural network · Graph neural network · xLSTM · Cooperative intelligent transportation system · Smart city

1 Introduction

The rapid growth of urban populations and vehicle usage has led to increasing congestion in metropolitan transportation networks, resulting in significant economic, environmental, and societal impacts. The global population exceeded 8 billion in 2022, with approximately 57% living in urban areas, and this proportion continues to grow [1]. At the same time, the number of vehicles on European roads has surpassed

252 million, placing increasing pressure on already constrained infrastructure [2]. As a result, traffic congestion has become a pervasive challenge, leading to substantial delays, economic losses, and negative health effects. For instance, an average European commuter loses approximately 90 hours per year due to congestion, while the economic cost of congestion has been estimated at tens of billions of dollars annually [3–5]. These trends highlight the urgent need for intelligent traffic management solutions to mitigate congestion and its wide-ranging impacts. One such approach is smart traffic management enabled by Cooperative Intelligent Transportation Systems (C-ITS), which can significantly alleviate road congestion.

The goal of C-ITS is to build smart services utilizing the communication, data collection and processing capabilities of participants in transportation and the infrastructure to make traffic more efficient and safe. Accurate short-term traffic forecasting is a key component of such applications,

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as it enables proactive traffic management strategies such as adaptive signal control and dynamic routing [6, 7]. These applications rely on models that can effectively capture both the temporal evolution of traffic conditions and the spatial dependencies imposed by the underlying road network.

Early approaches addressed only the temporal dimension, relying on recurrent architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), which capture sequential dependencies but ignore the spatial interactions between road segments. More recent work has turned to hybrid spatio-temporal architectures that represent the traffic network as a graph, enabling spatial correlations between road segments to be learned. Depending on how they model the temporal dimension, these approaches fall broadly into recurrent, convolutional, and attention-based families.

While these models substantially improved predictive accuracy, they share a common limitation on the temporal axis. Recurrent hybrids based on conventional LSTM or GRU units rely on scalar memory representations, which may limit their ability to retain heterogeneous long-range temporal dependencies. Fully convolutional models gain computational efficiency, but they capture long-range temporal structure only indirectly through stacked receptive fields. Attention-based models relate time steps flexibly, often with increased computational cost due to pairwise attention operations, but do not provide an explicit recurrent memory mechanism for preserving long-term temporal structure. These limitations are particularly consequential in urban traffic, where short-term signal-cycle fluctuations, medium-term peak-hour build-up, and longer-term effects such as congestion propagation and delayed dissipation coexist. Capturing these multi-timescale patterns simultaneously requires a temporal mechanism with richer and more structured memory, while retaining the graph-based spatial modeling established by previous spatio-temporal forecasting methods.

To address this gap, we propose TETRA, a hybrid spatio-temporal forecasting model that combines graph convolutional spatial modeling with Extended Long Short-Term Memory (xLSTM) for temporal modeling. xLSTM augments conventional recurrent architectures with stabilized exponential gating and a matrix-valued memory, yielding a higher-capacity temporal representation than scalar-memory recurrent units. This integration of structured recurrent memory with graph-based spatial modeling remains underexplored for traffic forecasting. This enhanced temporal memory is intended to support the modeling of multi-timescale traffic dynamics, including recurring peak patterns, congestion propagation, and delayed recovery.

We evaluate TETRA on two datasets: a real-world urban dataset collected in Istanbul (TRALICO) and the widely used METR-LA benchmark. To validate the advantages

of incorporating an xLSTM-based temporal mechanism into a spatio-temporal traffic flow prediction framework, we benchmark TETRA against strong, established, and widely used representatives of each main spatio-temporal paradigm: recurrent (T-GCN), convolutional (STGCN), and attention-based (GMAN) modeling. Extensive experiments, together with ablation, sensitivity, and statistical-significance analyses, show that TETRA outperforms or matches these representative baselines, with the most consistent gains at medium- and long-term horizons. On the real-world TRALICO dataset, it achieves MAE reductions of approximately 13% to 18% over the strongest representative baseline, depending on the prediction horizon.

The structure of this paper is organized as follows: Section 2 reviews commonly used models for traffic flow prediction. Section 3 describes the proposed TETRA framework and its architectural components. Section 4 presents the experimental results and compares TETRA with established spatio-temporal baseline models on benchmark datasets. Finally, Section 5 concludes the paper and outlines directions for future research.

2 Related Works

2.1 Temporal Traffic Prediction Approaches

Recurrent Neural Networks (RNNs) are frequently employed for traffic flow prediction due to their inherent ability to capture temporal dependencies and data trends. Generally, RNNs process time sequences $x_0, x_1, \dots, x_t$ by mapping them into hidden states $h_0, h_1, \dots, h_t$ using a set of trainable parameters Θ . The hidden state at time t is defined by $h_t = f(h_{t-1}, x_t, \Theta)$, creating a dependency chain where h_t relies on the preceding state h_{t-1} and input x_t . During training via backpropagation, gradients are often subject to the vanishing gradient problem. As derivatives are repeatedly multiplied, they can become negligibly small for distant timesteps, causing the model to ignore long-range dependencies.

To address these limitations, Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) were introduced. LSTM introduces gated memory mechanisms (input, forget, and output gates) to regulate information flow and mitigate vanishing gradients [8]. This architecture allows the model to efficiently store or discard information over long periods, making it a staple in traffic forecasting. Similarly, the GRU (2014) offers a streamlined alternative with only two gates, providing comparable performance with reduced computational overhead [9]. In their work, Tian et al. propose an *LSTM* modification which can effectively predict traffic even with non-complete training data by adding a masking

part to the gating mechanism [10]. Fu et al. evaluated *GRU* and *LSTM* on the same data set and claimed that although the performance of *GRU* is better, the improvement is not significant [11]. The most recent advancement in this lineage is Extended Long Short-Term Memory (*xLSTM*) to address the scalar bottleneck of traditional *LSTMs*. While standard *LSTMs* are limited by memory cells that store information in a single-scalar format, *xLSTM* introduces *sLSTM* and *mLSTM* blocks. The *sLSTM* utilizes exponential gating and a stabilizer state to handle non-linear surges in data, while *mLSTM* adopts a matrix-based memory to increase the representational capacity of the model, which may be beneficial for capturing complex temporal dependencies in traffic data. In their work, Alharthi et al. propose an *xLSTM*-based model for general time-series forecasting, reporting competitive results against established sequence-modeling benchmarks [12]. However, such applications treat the input as plain temporal sequences and do not exploit the spatial structure of road networks; the integration of *xLSTM* with graph-based spatial modeling therefore remains underexplored.

2.2 Spatio-Temporal Traffic Prediction Approaches

Accurate traffic forecasting requires the joint modeling of temporal dynamics and spatial dependencies across road networks. While purely temporal models, such as *RNNs*, effectively capture sequential patterns, they do not inherently account for spatial correlations between interconnected road segments. In transportation systems, traffic conditions propagate through the network, making spatial context a critical component for reliable prediction.

Early approaches addressed this limitation by the combination of Convolutional Neural Networks (*CNNs*) with *RNN*-based architectures [13]. In these models, *CNNs* are used to extract spatial features, while recurrent networks capture temporal evolution. While this approach seems promising, *CNNs* have their limitations as they operate on grid-structured data, whereas road networks typically exhibit non-Euclidean topology. To overcome this limitation, Graph Convolutional Networks (*GCNs*) were introduced, enabling direct modeling of spatial dependencies based on graph structures.

As illustrated in Fig. 1, *GCNs* generalize conventional convolution by aggregating information from neighboring nodes defined by an adjacency matrix, rather than relying on fixed grid neighborhoods. This allows traffic models to explicitly incorporate road connectivity, where the representation of each segment is enriched by information from its spatial neighbors. A comprehensive overview of convolutional architectures, including their graph-based extensions, is provided in [14].

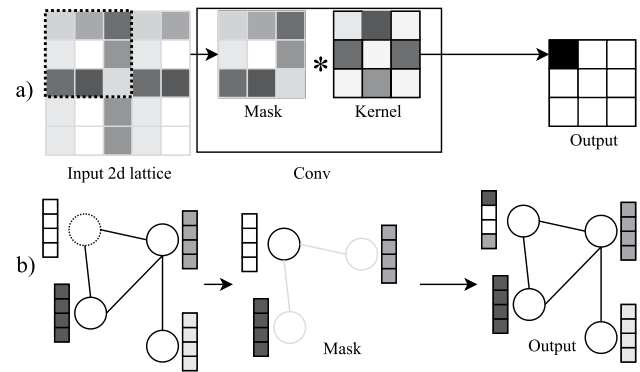


Fig. 1 The difference between 2D Convolutional Layer (a) and Graph Convolutional Layer (b)

Building on this foundation, a large body of work has focused on hybrid spatio-temporal architectures that pair *GCNs* with a temporal model, and these can be grouped by how they handle the temporal dimension into recurrent, convolutional, and attention-based families. The first family couples graph convolution with *recurrent* temporal models. One representative approach is *GCN-LSTM*, which integrates graph convolution with *LSTM* networks to jointly model spatial and temporal dependencies [15]. Similarly, the Temporal Graph Convolutional Network (*T-GCN*) combines *GCN* with *GRU*, where *GCN* captures spatial interactions and *GRU* models temporal dynamics [16]. While these recurrent architectures are effective for sequence modeling, their compressed hidden-state representations may limit their ability to preserve long-term temporal dependencies, particularly in complex urban traffic scenarios.

A second family replaces recurrence with *convolutional* temporal modeling. The Spatio-Temporal Graph Convolutional Network (*STGCN*) replaces recurrent networks with temporal convolutions, enabling parallel computation and improved efficiency [17], but purely convolutional temporal modeling may be less effective in capturing long-range dependencies compared to memory-based recurrent approaches.

A third family models temporal dependencies through *attention* mechanisms. The Graph Multi-Attention Network (*GMAN*) employs an encoder-decoder framework with multiple spatio-temporal attention blocks, allowing the model to directly relate historical observations to future time steps and reduce error accumulation in long-term forecasting [18]. Building on this attention-based formulation, more recent work refines it with Transformer-inspired designs and adaptive or multi-scale spatial modeling: the Spatial-Temporal Graph Transformer Network (*STGTN*) combines causal convolutions with masked graph attention [19]. Similarly, *ST-TransNet* introduces incremental positional embeddings and multi-head self-attention to capture dynamic patterns and external influences such as holiday effects [20]; and

the Long-Term Spatio-Temporal Graph Attention Network (*LSTGAN*) integrates convolutional downsampling, graph convolution, and self-attention to model multi-scale dependencies across space and time [21]. Across these works, attention relates time steps flexibly but without an explicit recurrent memory for long-term temporal structure, and at a cost that grows with sequence length.

Overall, the literature shows a clear progression across the three temporal families, from recurrent and convolutional hybrids toward attention-driven architectures. Yet a common limitation persists on the temporal axis: recurrent hybrids compress heterogeneous temporal dynamics into a scalar hidden state, convolutional models capture long-range structure only indirectly, through stacked receptive fields, and attention-based models, while flexible, lack an explicit recurrent memory for long-term dependencies and scale with the number of pairwise interactions. Across all three families, the spatial side has matured around graph convolution, but few widely used approaches pair graph-based spatial modeling with a structured, high-capacity recurrent memory on the temporal side. This gap motivates combining explicit graph-based spatial modeling with a more expressive temporal memory mechanism. We address it by introducing TETRA, a hybrid framework that integrates graph convolution with an xLSTM-based temporal module, aiming to improve the modeling of long-term temporal dependencies while preserving the structural properties of traffic networks.

3 Methodology

The objective of the proposed model is to predict road-related traffic variables (e.g., mean speed, occupancy or traffic flow) using road topology and historical observations. The road network is represented at time t as a graph $G_t = (V_t, E)$, where $V_t = [v_{0,t}, \dots, v_{n,t}]$ denotes the set of nodes corresponding to n road segments, and each node feature $v_{i,t} \in \mathbb{R}^f$ encodes f traffic attributes of segment i at time t . Each node may contain multiple traffic attributes (e.g., flow and mean speed), in which case $f > 1$. If only a single attribute is available, then $v_{i,t} \in \mathbb{R}$. E denotes the set of edges, representing connectivity between road segments. Connectivity information is stored in the adjacency matrix $A \in \mathbb{R}^{n \times n}$:

$$a_{ij} = \begin{cases} 1 & \text{if there is } v_i \rightarrow v_j \text{ edge or } i = j \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

To incorporate spatial proximity, we define a distance-based weight matrix $S \in \mathbb{R}^{n \times n}$, where $s_{ij} = \frac{1}{\text{distance}(v_i, v_j)}$ for connected nodes (otherwise $s_{i,j} = 0$), assigning greater importance to closer neighbors.

The final weighted adjacency matrix is $A' = A \odot (S + I_n)$. Note that A' is fixed and shared across all timesteps, reflecting the assumption of a static road-network topology over the forecasting window.

We formalize the traffic prediction model as follows:

$$[\hat{V}_{t+1}, \dots, \hat{V}_{t+p}] = f([V_{t-h}, \dots, V_t], A', W, B) \quad (2)$$

$[\hat{V}_{t+1}, \dots, \hat{V}_{t+p}]$ contains the estimated future road information from timestep $t+1$ to timestep $t+p$, where p is the length of the prediction horizon, h is the historical window size, and W, B are trainable parameters of model f . The objective is to learn model parameters W and B that minimize the prediction error as seen in Eq. 3, where $\hat{V}_{t+i}$ and V_{t+i} denote the predicted and ground truth values at future timestep $t+i$ respectively.

$$\min_{W, B} \text{loss}([V_{t+1}, \dots, V_{t+p}], [\hat{V}_{t+1}, \dots, \hat{V}_{t+p}]) \quad (3)$$

3.1 TETRA Architecture

In this paper we propose TETRA, a hybrid spatio-temporal model that combines graph convolutional spatial feature extraction with an xLSTM-based recurrent temporal module. As shown in Fig. 2, the architecture consists of four main stages: spatial feature extraction using GCNs, projection into a compact latent representation, temporal modeling using xLSTM blocks, and final prediction through a fully connected layer. The model takes a sequence of graph snapshots over a historical window of length h , where each snapshot encodes traffic features for all nodes in the network. Each observation corresponds to a graph $G_t = (V_t, E)$, resulting in an input tensor $X \in \mathbb{R}^{h \times n \times f}$, where n is the number of nodes and f is the number of features per node.

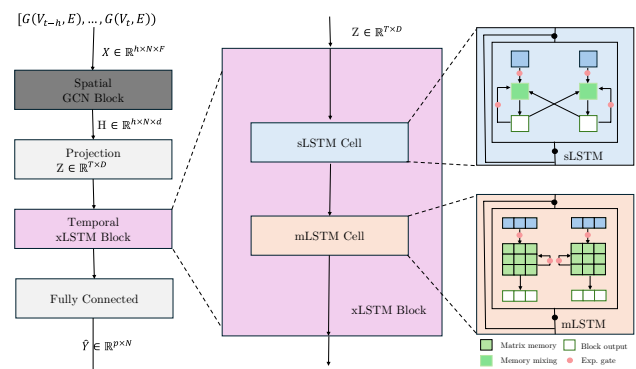


Fig. 2 Architecture of the proposed model, TETRA to capture spatio-temporal dependencies on graph-like data. The left part shows the end-to-end processing pipeline, while the right part details the internal structure of the xLSTM block, including the sLSTM and mLSTM components

Spatial Modeling Spatial dependencies are captured using GCN. This operation aggregates information from neighboring nodes according to the weighted adjacency matrix defined in Eq. 1. The same GCN parameters are shared across all timesteps, ensuring consistent spatial feature extraction throughout the sequence. The propagation rule of a GCN layer is given by:

$$H^{(k+1)} = \psi(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(k)} W^{(k)}), \quad (4)$$

where $\tilde{A}$ is the normalized weighted adjacency matrix A' with self-loops, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$ is the degree matrix, $W^{(k)}$ are trainable weights, and $\psi(\cdot)$ denotes a nonlinear activation function. As a result, the representation of each node is enriched with contextual information from its local neighborhood, producing a transformed tensor $H \in \mathbb{R}^{h \times n \times d}$, where d is the embedding dimension of the output of the GCN layer. Following spatial feature extraction, a projection layer maps node-level representations into a compact latent sequence representation $Z \in \mathbb{R}^{h \times D}$. This step reduces dimensionality and enables efficient temporal modeling by transforming the graph-structured data into a sequence format. The number of GCN layers is a hyperparameter of the model.

Temporal Modeling To capture long-term temporal dependencies, TETRA employs an xLSTM block. xLSTM extends conventional recurrent architectures by incorporating both scalar and matrix-based memory mechanisms (shown in Fig. 2). In a standard LSTM, the hidden state is controlled by input, forget, and output gates:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (5)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (6)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (7)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t), \quad (8)$$

where c_t denotes the cell state, h_t the hidden state and $\tilde{c}_t$ the candidate cell state used to update the memory through the input gate at time step t .

The sLSTM cell extends this formulation by replacing standard sigmoid gating with stabilized exponential gating:

$$z_t = W_z x_t + U_z h_{t-1} + b_z, \quad (9)$$

$$i_t = \exp(W_i x_t + U_i h_{t-1} + b_i) \quad (10)$$

$$f_t = \exp(W_f x_t + U_f h_{t-1} + b_f), \quad (11)$$

$$m_t = \max(\log f_t + m_{t-1}, \log i_t), \quad (12)$$

$$\tilde{i}_t = \exp(\log i_t - m_t) \quad (13)$$

$$\tilde{f}_t = \exp(\log f_t + m_{t-1} - m_t), \quad (14)$$

$$c_t = \tilde{f}_t \odot c_{t-1} + \tilde{i}_t \odot z_t. \quad (15)$$

m_t is a stabilizer state that improves numerical robustness when using exponential gates.

mLSTM replaces the conventional cell state with a matrix-valued memory:

$$q_t = W_q x_t, \quad k_t = W_k x_t, \quad v_t = W_v x_t, \quad (16)$$

$$i_t = \exp(W_i x_t + U_i h_{t-1} + b_i) \quad (17)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \quad (18)$$

$$C_t = f_t C_{t-1} + i_t v_t k_t^\top, \quad (19)$$

where $C_t \in \mathbb{R}^{d_m \times d_m}$ denotes the matrix memory state. The hidden representation is then obtained by querying the memory through q_t , enabling the modeling of higher-order temporal interactions that scalar memory cannot capture.

The sLSTM and mLSTM components serve complementary roles in temporal modeling. sLSTM extends conventional LSTM by improving numerical stability and long-term memory retention through exponential gating and normalization, making it well-suited for capturing persistent temporal patterns in traffic data. mLSTM introduces a matrix-valued memory representation, enabling the model to encode and retrieve high-order interactions between temporal features that cannot be represented by scalar memory alone. The advantages of matrix-based memory are particularly relevant for traffic forecasting tasks. Urban traffic exhibits strong multi-timescale temporal dynamics, including short-term fluctuations (e.g., signal cycles), medium-term patterns (e.g., peak-hour buildup), and long-term dependencies such as congestion propagation and delayed dissipation effects. Standard LSTM architectures rely on scalar memory representations, which compress these heterogeneous temporal dependencies into a limited latent space, potentially leading to information loss over extended horizons. The matrix memory of mLSTM enables the model to retain and update multiple interacting temporal patterns simultaneously, allowing it to capture complex relationships such as recurring peak structures, spillback effects, and delayed recovery dynamics.

By combining these two mechanisms, the xLSTM block is able to simultaneously preserve stable long-term dependencies and model complex multi-dimensional temporal relationships. This is particularly important in traffic forecasting, where congestion patterns evolve over extended

time horizons while exhibiting structured dependencies across multiple features and locations.

4 Evaluation

This section evaluates the proposed TETRA model in terms of prediction accuracy, robustness, and computational performance. The evaluation is conducted on two datasets and includes comparisons with established baseline models representative of the main spatio-temporal paradigms under identical experimental settings. Figure 3 illustrates the construction of graph-structured inputs from raw traffic sensor location data. Figure 3(a) shows the spatial layout of traffic sensors deployed on a road network. In Fig. 3(b), this layout is converted into a graph representation, where each node corresponds to a sensor location and edges encode spatial relationships between sensors, including their physical distances. Traffic measurements (e.g., speed or volume) at time t are represented as node features V_t . Based on this representation, the adjacency matrix A and the distance-based weight matrix S are constructed, from which the weighted adjacency matrix $A' = A \odot (S + I_n)$ is derived, as illustrated in Fig. 3(c). This transformation enables the incorporation of spatial relationships between sensors into the learning process. It is important to note that A' remains constant over time, reflecting the assumption of a static road network topology. This assumption is reasonable for short-term traffic forecasting, where the physical road topology remains constant and dynamic changes are primarily reflected in node features rather than graph structure.

To evaluate the performance of the proposed model and compare it with established baseline approaches, two datasets are employed (See Table 1):

Table 1 Summary of the METR-LA and TRALICO datasets

Property	METR-LA	TRALICO
Data Type	Speed	Volume
#Sensors	207	54
#Timesteps	34,271	61,056
Interval	5 min	5 min
Duration	~119 d	~212 d
Area	Highway net.	5 urban intersec.
Granularity	Road seg.	Lane-level
Mean	53.72 km/h	265.85 veh/hour
Std	20.26	364.87
Range	[0, 70]	[0, 2267.7]

METR-LA is a publicly available benchmark dataset for spatio-temporal traffic forecasting. It contains approximately four months of traffic data collected in Los Angeles at 5-minute intervals. The dataset provides travel speed measurements from 207 loop detectors deployed across the road network.

Istanbul dataset (TRALICO) is provided by the TRALICO project and covers a central urban area in Istanbul, Turkey. In total, five interconnected intersections were monitored, with lane-level traffic measurements collected at each location (see Fig. 4). The sensors record traffic volume with irregular sampling intervals. For this study, approximately seven months of data were selected and preprocessed. First, measurements were standardized and resampled to fixed 5-minute intervals. Missing values were imputed using linear interpolation. To mitigate the impact of sensor errors, extreme values exceeding the 99th percentile were capped using a sensor-specific thresholding strategy. Additionally, sinusoidal and cosinusoidal time encoding features were added to capture periodic temporal patterns such as time-of-day and day-of-week effects. For both datasets, 70% of the data is used for training, 15% for validation, and 15% for testing.

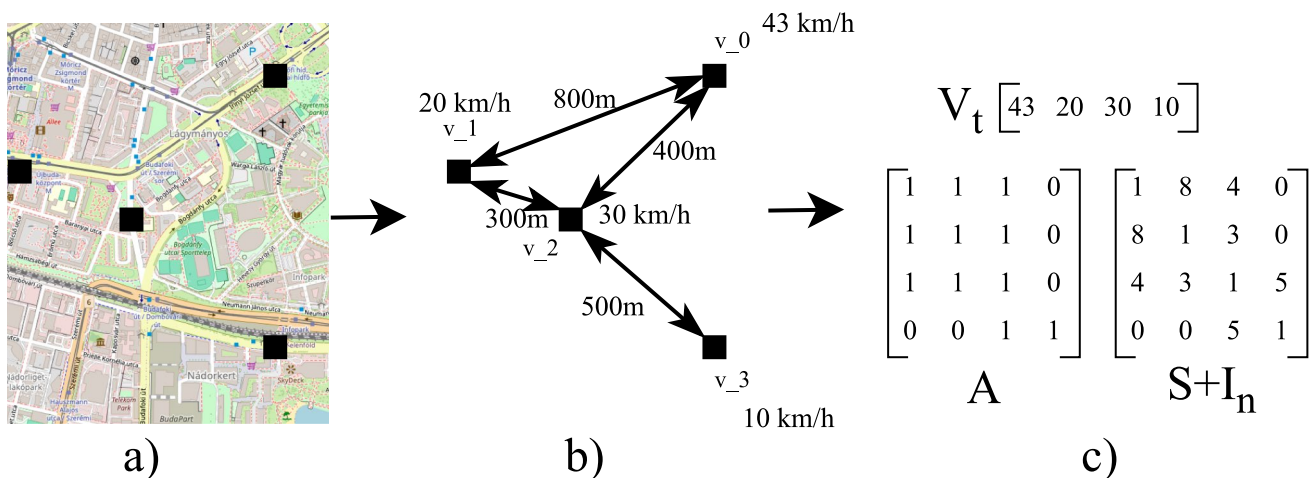


Fig. 3 Transformation of raw traffic sensor data into graph-structured input: (a) sensor layout, (b) graph representation with spatially weighted edges, (c) derived matrices and node features V_t

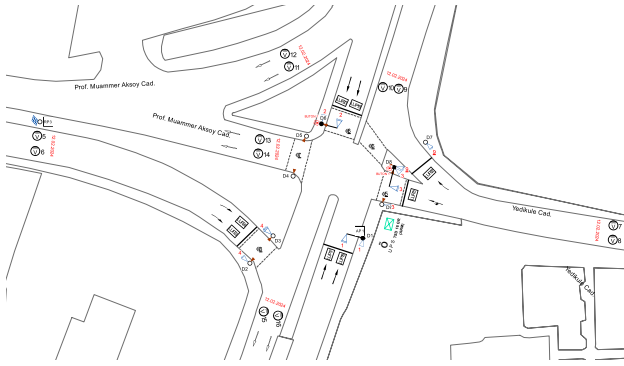


Fig. 4 Sensor topology for one of the monitored junctions in Istanbul. “V” and “LUP” denote loop detectors monitoring volume on a particular lane. Volume is defined as the number of vehicles passing through the detector during the sampling phase

4.1 Evaluation Metrics

The performance of the models is evaluated using standard regression metrics commonly used in traffic forecasting.

Mean Absolute Error (MAE) is used as the primary evaluation metric:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (20)$$

Additionally, Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2) are reported to provide complementary perspectives on prediction accuracy. RMSE (21) penalizes larger errors better than MAE (20), because of the squaring function. Conversely, MAE is easier to interpret since it retains the original unit of the data. R^2 (22) indicates how well the model can explain the variance of the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (21)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (22)$$

4.2 Baseline Models

To evaluate the effectiveness of the proposed TETRA model, we compare it against several well-established and highly cited models used in spatio-temporal traffic forecasting. We select T-GCN, STGCN, and GMAN as baselines because together they represent the three dominant

temporal-modeling paradigms in graph-based traffic forecasting — recurrent (T-GCN), convolutional (STGCN), and attention-based (GMAN). All three operate on a pre-defined adjacency derived from network topology and sensor distances rather than a learned adaptive graph, and rely solely on historical traffic and network structure without exogenous covariates. All baselines provide publicly available implementations, enabling a fair and reproducible comparison under identical settings.

All models are trained and evaluated under a unified computational environment to ensure a fair and reproducible comparison. The experiments were conducted on a system equipped with an NVIDIA L40S GPU (48 GB VRAM), 188 GB system memory, and 16 vCPUs. Model implementations follow their original configurations where applicable, with hyperparameters tuned consistently across models when necessary.

Training is performed using the Adam optimizer with an initial learning rate of 0.0001. Early stopping is applied based on validation loss with a patience of 20 epochs to prevent overfitting. The batch size is set to 256 for all models unless otherwise specified. All input features are normalized prior to training using standard scaling to improve convergence stability.

4.3 Hyperparameter Configuration

Hyperparameters are optimized using GridSearchCV on the TRALICO dataset, and the selected configuration is applied to the METR-LA dataset. The selected hyperparameters are listed in Table 2. For training we used early stopping (delta = 0, patience = 20), ADAM optimizer and MAE as loss function. We evaluate the models on prediction horizons of $T = 3, 6, 12$, corresponding to 15, 30, and 60 minutes.

Table 2 Hyperparameter options used for GridSearchCV to optimize TETRA and the final selected values

Hyperparameter	Value space	Chosen value
GCN channel projection	[1, 2, 4, 8, 16, 32, 64]	32
Number of GCN layers	[1, ..., 10]	1
Embedding dimension	[128, 256, 512, 1024]	256
xLSTM block stack configuration	[[sLSTM], [mLSTM], [sLSTM, mLSTM], 2 × [sLSTM, mLSTM]]	[sLSTM, mLSTM]
mLSTM kernel, QKV size, number of blocks	[[2, 1, 1], [4, 2, 2], [8, 4, 4]]	[4, 2, 2]
sLSTM number of heads, kernel size	[[1, 1], [2, 2], [4, 4]]	[2, 2]
sLSTM projection factor	[1.0, 1.1, 1.2, 1.3, 1.4, 1.5]	1.1
Batch size	[16, 32, 128, 256, 512]	256

4.4 Predictive Performance

Tables 3 and 4 show the predictive performance of the evaluated models on the TRALICO and METR-LA datasets, across short-, medium-, and long-term forecasting horizons ($p \in 3, 6, 12$).

On the TRALICO dataset, TETRA consistently achieves the best performance across all prediction horizons and evaluation metrics. For short-term prediction ($p = 3$), TETRA attains the lowest MAE (60.8) and RMSE (111.8), along with the highest R^2 score (0.90); compared to the strongest baseline, STGCN, this corresponds to a relative MAE reduction of approximately 13%. As the prediction horizon increases, all models exhibit performance degradation, but TETRA maintains a clear margin over the baselines. At $p = 6$, it reduces MAE by approximately 13% relative to STGCN and 10% relative to GMAN, while improving RMSE by 11% and 23% over STGCN and GMAN, respectively. This advantage becomes even more pronounced at

Table 3 Results on the TRALICO dataset. * indicates statistically significant improvement over the best baseline ($p < 0.05$, two-sample t-test, $n = 5$)

p	Model	MAE	RMSE	R^2
3	GMAN	70.6 ± 1.5	140.8 ± 3.0	0.85 ± 0.01
	STGCN	69.9 ± 0.0	120.0 ± 0.4	0.89 ± 0.01
	T-GCN	93.2 ± 0.2	155.4 ± 1.2	0.81 ± 0.01
	TETRA	$60.8 \pm 0.2^*$	$111.8 \pm 0.4^*$	$0.90 \pm 0.01^*$
6	GMAN	74.2 ± 0.3	151.1 ± 1.2	0.83 ± 0.01
	STGCN	76.3 ± 0.1	130.8 ± 1.1	0.87 ± 0.01
	T-GCN	95.5 ± 0.3	159.7 ± 0.2	0.80 ± 0.01
	TETRA	$66.7 \pm 0.2^*$	$117.0 \pm 0.4^*$	$0.89 \pm 0.01^*$
12	GMAN	76.6 ± 0.9	155.0 ± 3.0	0.82 ± 0.01
	STGCN	85.9 ± 0.1	174.5 ± 1.4	0.83 ± 0.01
	T-GCN	101.3 ± 0.3	168.9 ± 0.7	0.78 ± 0.01
	TETRA	$70.3 \pm 0.1^*$	$124.0 \pm 0.3^*$	$0.88 \pm 0.01^*$

Table 4 Results on the METR-LA dataset. * indicates statistically significant improvement over the best baseline ($p < 0.05$, two-sample t-test, $n = 5$)

p	Model	MAE	RMSE	R^2
3	GMAN	2.9 ± 0.1	5.7 ± 0.2	0.82 ± 0.02
	STGCN	2.8 ± 0.1	5.4 ± 0.1	0.84 ± 0.00
	T-GCN	3.1 ± 0.1	5.8 ± 0.2	0.82 ± 0.01
	TETRA	2.8 ± 0.1	5.5 ± 0.2	0.89 ± 0.01
6	GMAN	3.2 ± 0.1	6.7 ± 0.2	0.74 ± 0.02
	STGCN	3.2 ± 0.1	6.5 ± 0.2	0.76 ± 0.01
	T-GCN	3.6 ± 0.1	6.9 ± 0.2	0.73 ± 0.01
	TETRA	$3.0 \pm 0.1^*$	$6.0 \pm 0.2^*$	$0.78 \pm 0.01^*$
12	GMAN	3.5 ± 0.1	7.5 ± 0.1	0.65 ± 0.02
	STGCN	3.8 ± 0.0	7.8 ± 0.2	0.64 ± 0.00
	T-GCN	4.3 ± 0.1	8.4 ± 0.2	0.60 ± 0.01
	TETRA	$3.3 \pm 0.1^*$	$6.3 \pm 0.2^*$	$0.69 \pm 0.01^*$

the longest horizon ($p = 12$), where TETRA outperforms STGCN by nearly 18% and GMAN by 8% in MAE, and yields a substantial RMSE reduction of 29% over STGCN and 20% over GMAN. To assess whether these improvements are statistically significant, we performed two-sample t-tests across five runs.

The results confirm that TETRA significantly ($p < 0.05$) outperforms the baseline methods on TRALICO across all prediction horizons. Additionally, TETRA exhibits consistently low standard deviations, indicating stable convergence and reliable generalization across runs.

On the METR-LA dataset, TETRA achieves the best or most competitive performance across all prediction horizons, with its advantage growing as the horizon increases. For short-term prediction ($p = 3$), TETRA matches the strongest baseline in MAE (2.8) and is marginally behind STGCN in RMSE (5.5 vs. 5.4), while achieving a notably higher R^2 score (0.89 vs. 0.84), indicating superior explanatory power even where the error metrics are comparable. For medium- and long-term horizons, TETRA consistently outperforms all baselines across every metric. At $p = 6$, it reduces MAE by approximately 6% relative to both STGCN and GMAN, while improving RMSE by 8% and 10% over STGCN and GMAN, respectively. This improvement becomes even more pronounced at the longest horizon ($p = 12$), where TETRA yields a reduction of approximately 6% in MAE and 16% in RMSE, firmly outperforming GMAN across both metrics. Statistical testing confirms that these improvements are statistically significant ($p < 0.05$) for the medium- and long-term horizons. For short-term prediction ($p = 3$), where the differences in MAE and RMSE are minimal, the gains in those specific metrics do not reach statistical significance; however, the improvement in R^2 remains consistent across all horizons. Specifically, TETRA yields a relative R^2 increase over the strongest baseline of 6% at $p = 3$, 2.6% at $p = 6$, and 6.2% at $p = 12$, demonstrating more robust explanatory power even when error metrics are close. This pattern mirrors the TRALICO results: TETRA is most beneficial at medium and long horizons, where its structured temporal memory helps preserve long-range dependencies that the baselines capture less effectively.

Figure 5 shows a comparison between the ground truth and the predicted volume of TETRA on a selected road segment of the TRALICO dataset. Figure 5a shows results for a 1-day window, Fig. 5b depicts predictions over a 3-day window. In both cases, the model is able to capture the overall temporal dynamics, including peak traffic periods and recovery phases, with predictions closely following the ground truth. Although some errors appear during sudden fluctuations, the results indicate that TETRA generalizes well across both short and extended time horizons.

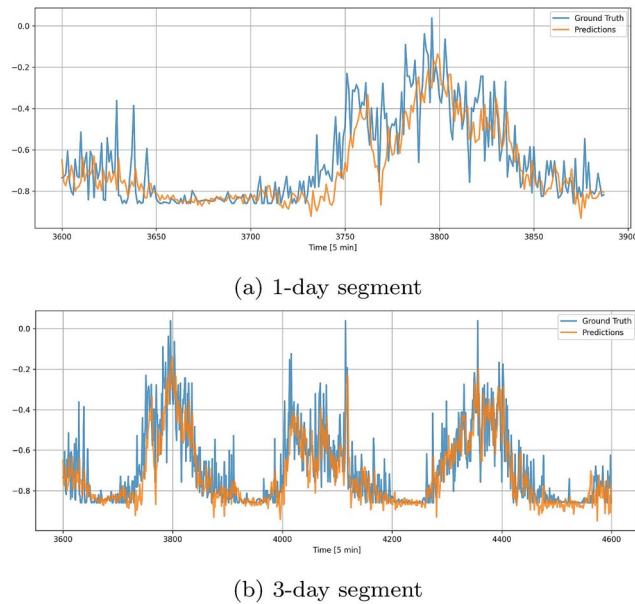


Fig. 5 Predicted volume by TETRA compared with ground truth on TRALICO for 1-day (a) and 3-day (b) segments, with prediction horizon $p = 12$

We examined the distribution of prediction errors and their relationship with traffic volume. Figure 6 presents the distribution of absolute errors across all prediction horizons $p \in [1, 12]$ on TRALICO dataset. The distribution is heavily right-skewed, with the majority of errors concentrated at lower values and a long tail corresponding to rare but large deviations. This indicates that the model performs reliably under typical traffic conditions, while extreme errors occur infrequently, likely corresponding to abrupt traffic changes or anomalous events.

Figure 7 illustrates the relationship between traffic volume and mean absolute error. A clear monotonic trend can be observed, where prediction error increases with traffic

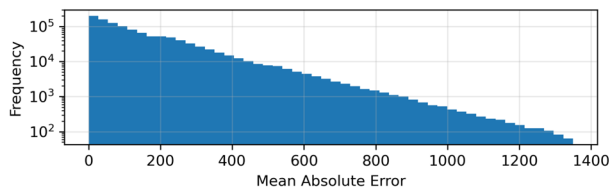


Fig. 6 Error distribution of TETRA on TRALICO dataset across all the $p \in [1, 12]$ observed horizon

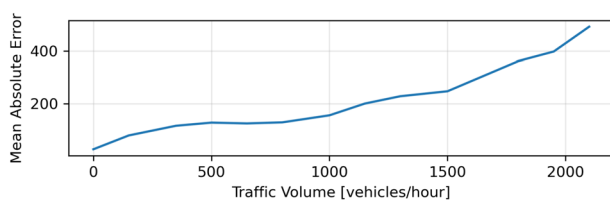


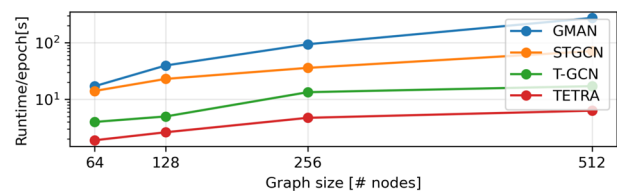
Fig. 7 Relationship between average MAE and traffic volume on TRALICO dataset

volume. At low traffic levels, the model achieves relatively low errors, reflecting the more stable and predictable dynamics of uncongested conditions. However, as volume increases, the system becomes more nonlinear and susceptible to congestion effects, leading to higher prediction uncertainty and larger errors. Notably, the error growth accelerates in high-volume regimes, suggesting that congestion propagation and saturation effects introduce additional modeling challenges. This analysis also suggests that incorporating additional contextual information (e.g., incidents or control signals) could further improve performance in high-volume scenarios.

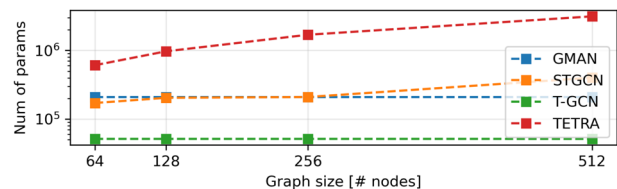
Figure 8 compares runtime per epoch (Fig. 8a) and parameter scaling (Fig. 8b) as the graph size n in $X \in \mathbb{R}^{h \times n \times f}$ increases. TETRA shows moderate runtime growth, increasing from 1.9 s at 64 nodes to 6.32 s at 512 nodes, while remaining significantly faster than GMAN and STGCN at larger scales. In contrast, GMAN exhibits the steepest runtime increase, reaching 273.6 s per epoch, indicating limited scalability despite its fixed parameter count. Although TETRA has a higher number of parameters, its runtime remains efficient. This is because its complexity is dominated by dense linear and recurrent operations, which are highly optimized and scale well on modern hardware. In contrast, attention-based models (such as GMAN) incur quadratic complexity due to pairwise node interactions, leading to significantly higher computational cost. Additionally, TETRA's graph convolution operates on sparse structures, further improving efficiency. These results show that TETRA achieves a favorable trade-off between model capacity and scalability.

4.5 Ablation

To systematically evaluate the contribution of each architectural component, we design the ablation study to reflect



(a) Runtime (seconds per Epoch)



(b) Number of Parameters

Fig. 8 Scalability comparison on TRALICO dataset

four fundamental modeling paradigms commonly used in traffic forecasting:

- temporal-only models, which capture sequential dynamics without spatial context (e.g., sLSTM, mLSTM, xLSTM)
- spatial-only models, which model network structure without temporal recurrence (e.g., GCN)
- simple hybrid models, which combine spatial and temporal modeling using conventional recurrent units (e.g., GCN + LSTM)
- the proposed advanced hybrid model **TETRA**, which integrates graph-based spatial modeling with enhanced temporal memory mechanisms.

This decomposition allows for a direct comparison between traditional approaches and the proposed architecture, isolating the impact of both spatial modeling and advanced temporal memory.

TETRA consistently achieves the lowest MAE across all prediction horizons ($p=3,6,12$), with the performance gap increasing for longer horizons (See Fig. 9). While all components degrade as the horizon grows, the full model exhibits a slower degradation, indicating improved long-range temporal modeling. In terms of R^2 TETRA remains competitive and stable across all settings. The results clearly reflect the hierarchical contribution of the modeling paradigms. Temporal-only models fail to capture spatial propagation effects, while spatial-only models lack the ability to

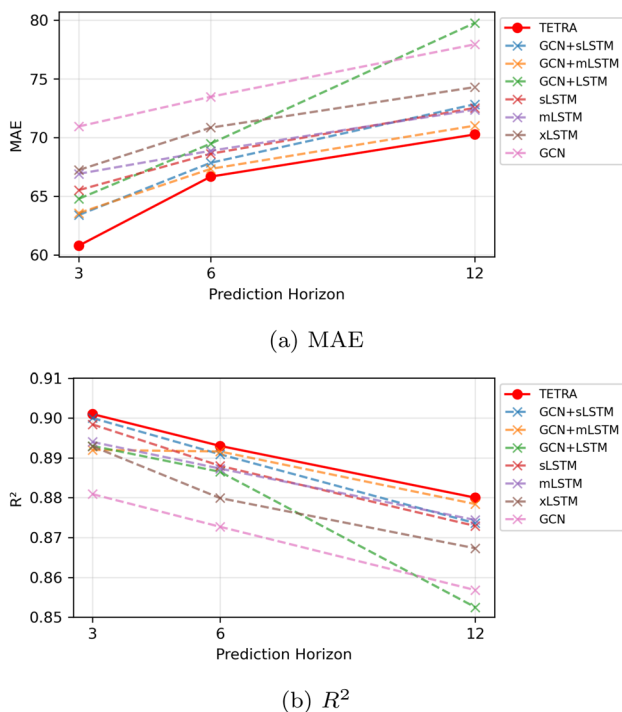


Fig. 9 Ablation study of TETRA on TRALICO dataset

model temporal evolution. Simple hybrid models improve performance by combining both aspects but remain limited in capturing long-term dependencies. In contrast, the full TETRA model consistently achieves the best performance, demonstrating that both spatial modeling and enhanced temporal memory are necessary to accurately capture complex traffic dynamics, particularly for longer prediction horizons.

4.6 Parameter Sensitivity

We evaluate the sensitivity of TETRA to key architectural and training hyperparameters, including the number of GCN layers, loopback window length, batch size and embedding dimension.

Increasing the **number of GCN layers** degrades performance across all prediction horizons. As shown in Fig. 10, MAE exhibits a clear upward trend with depth, with the best performance consistently achieved at shallow configurations (1–2 layers). This behavior is consistent with the well-known over-smoothing effect in graph convolutional networks, where node representations become increasingly indistinguishable as depth grows [22]. The results indicate that a shallow GCN is sufficient to capture the relevant spatial dependencies for this task, and that additional layers provide no benefit.

Figure 11 shows that increasing the **temporal input window** improves performance up to a certain point. Moving from short windows (3) to moderate lengths (6–12) yields consistent MAE reductions across all horizons, while further increase of the window size provides diminishing returns and may slightly degrade performance.

The **batch size** study (See Fig. 12) reveals a clear trade-off between performance and computational efficiency.

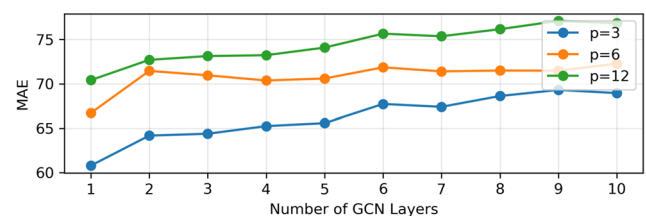


Fig. 10 Relationship between MAE and number of GCN layers on TRALICO dataset

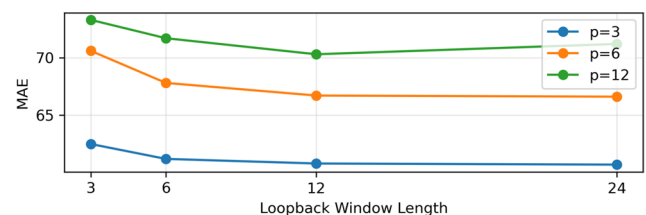


Fig. 11 Relationship between MAE and temporal input window (loopback) size h on TRALICO dataset

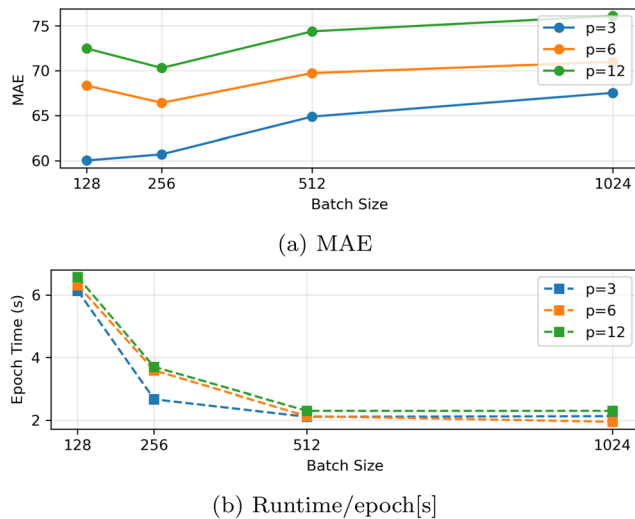


Fig. 12 Effect of batch size on TETRA

Smaller batch sizes (128–256) achieve the best accuracy, while larger batches (512–1024) lead to noticeable degradation, particularly for longer prediction horizons. At the same time, increasing batch size reduces training time per epoch. This indicates that moderate batch sizes (e.g., 256) provide a good balance between convergence quality and computational efficiency.

The embedding dimension has a pronounced effect on performance shown in Fig. 13. Increasing the dimension from 128 to 256 consistently improves MAE and R^2 across all prediction horizons. However, further increasing it to 512 does not yield additional gains and may slightly degrade performance while increasing runtime. This indicates that an embedding size of 256 is sufficient to capture the relevant spatio-temporal patterns without overparameterization.

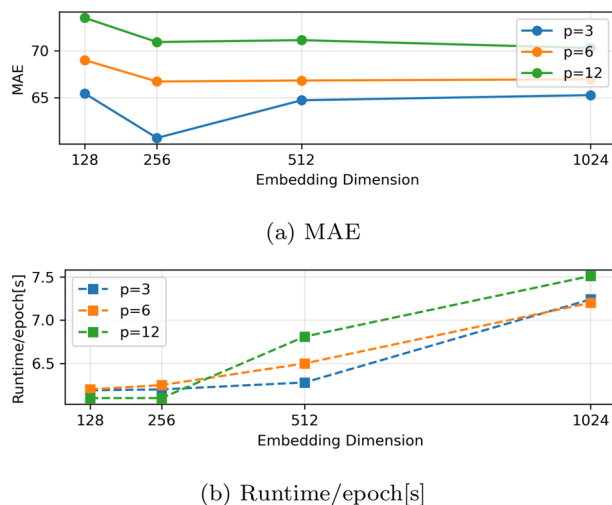


Fig. 13 Effect of embedding dimension size on TETRA

4.7 Data Preprocessing

We evaluate the impact of key preprocessing steps by modifying the baseline pipeline (A0/B0/C0) along three dimensions: imputation strategy (A), outlier capping (B), and cyclical encoding (C) (See Fig. 14).

Linear interpolation (A0) yields the best results, while forward fill (A1) degrades performance and hourly aggregation (A2) results in a modest increase in error. The choice of imputation method has a small but consistent impact.

Outlier handling also has a pronounced effect: disabling capping (B1) significantly worsens performance, while a stricter threshold (B2, 95th percentile) performs slightly better than the baseline, confirming that the 99th percentile provides a better trade-off between noise reduction and information retention.

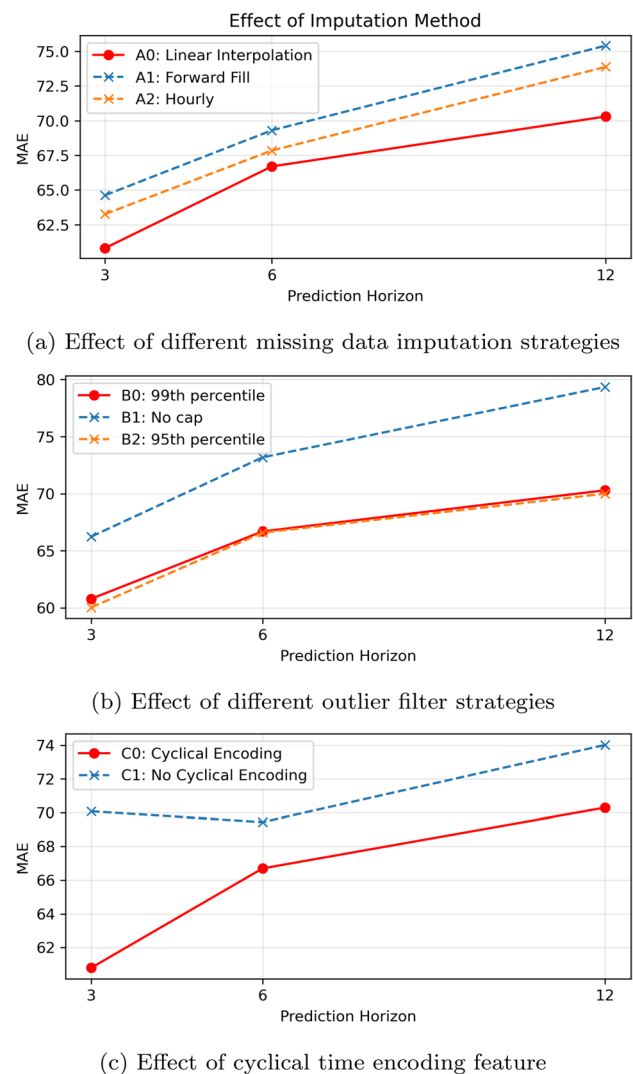


Fig. 14 Data preprocessing sensitivity analysis on TRALICO dataset

Removing cyclical encoding (C1) leads to the largest degradation across all horizons, with MAE increasing substantially (e.g., +9.3 at $p = 3$), highlighting the importance of capturing periodic temporal patterns.

The results indicate that temporal feature engineering and proper outlier handling are critical for accurate forecasting, whereas imputation plays a secondary role.

5 Conclusion and Future Works

This paper introduced TETRA, a hybrid spatio-temporal traffic forecasting model that integrates Graph Convolutional Networks (GCNs) for spatial modeling with Extended Long Short-Term Memory (xLSTM) for temporal modeling. The architecture is designed to capture both spatial dependencies across road networks and complex temporal dynamics, including long-range dependencies that are difficult to model with conventional recurrent or convolutional approaches. We evaluated the model on two datasets: TRALICO, a real-world dataset collected in Istanbul, and METR-LA, a widely used benchmark, across multiple prediction horizons, and compared it against established baselines representative of the recurrent (T-GCN), convolutional (STGCN), and attention-based (GMAN) paradigms.

The empirical results show that TETRA consistently outperforms or matches these baselines, with the most pronounced gains observed at medium- and long-term horizons.

On the TRALICO dataset, TETRA achieves the lowest error metrics at the short-term horizon ($p = 3$), corresponding to a relative MAE reduction of 13.0% and an RMSE reduction of 6.8% over the strongest baseline, STGCN, while outperforming GMAN by 13.9% in MAE and 20.6% in RMSE. At the medium horizon ($p = 6$), TETRA reduces MAE by 12.6% relative to STGCN and 10.1% relative to GMAN, while improving RMSE by 10.6% and 22.6%, respectively. At the longest horizon ($p = 12$), the performance margins become even more pronounced, reaching an MAE reduction of 18.2% over STGCN and 8.2% over GMAN, alongside substantial RMSE reductions of 28.9% over STGCN and 20.0% over GMAN. Statistical significance analysis confirms that these improvements are highly robust across horizons on TRALICO.

On the METR-LA dataset, TETRA delivers highly competitive performance. For short-term forecasting ($p = 3$), it matches the strongest baseline in MAE and falls marginally behind STGCN in RMSE (5.5 vs. 5.4), while achieving a notably higher R^2 score (0.89 vs. 0.84), which indicates superior explanatory power. At longer horizons, it consistently outperforms all baselines across every metric. At

$p = 6$, TETRA reduces MAE by approximately 6% relative to both STGCN and GMAN, while improving RMSE by 8% over STGCN and 10% over GMAN. At $p = 12$, the improvement reaches approximately 6% in MAE and 16% in RMSE, clearly outperforming GMAN on both metrics. Statistical testing confirms that these medium- and long-term improvements are statistically significant ($p < 0.05$). While the short-term error differences on METR-LA do not reach statistical significance, the improvement in R^2 remains consistent across all horizons (yielding a relative increase over the strongest baseline of 6% at $p = 3$, 2.6% at $p = 6$, and 6.2% at $p = 12$).

The model also exhibits favorable scalability, maintaining competitive runtime despite a larger number of parameters. The ablation and sensitivity analyses provide further insight into these results, showing that both graph-based spatial modeling and structured temporal memory are necessary for accurate forecasting. In particular, the matrix-valued memory of xLSTM enables the model to capture multi-timescale traffic dynamics—including recurring peak patterns, congestion propagation, and delayed recovery—that scalar memory represents less effectively, which helps explain the model's advantage at longer horizons.

Despite these advantages, the current formulation has several limitations. First, the model relies on a static adjacency matrix, assuming that spatial dependencies between road segments remain constant over time; this simplification may limit its ability to capture dynamic or latent interactions arising from changing traffic conditions, incidents, or control interventions. Second, the experimental comparison is restricted to established representatives of the main temporal paradigms under a common predefined-graph setting, and does not include recent transformer-based or adaptive-graph models. Third, the model assumes full knowledge of the road network and requires centralized access to all input data, which may restrict applicability in large-scale or distributed environments.

Future work will address these limitations along three directions. First, incorporating adaptive or dynamic adjacency learning could enable the model to capture time-varying spatial dependencies and non-local interactions. Second, benchmarking TETRA against recent transformer-based and adaptive-graph models, ideally on larger road networks, would further situate its performance within the current state of the field. Third, investigating distributed or partitioned variants—where different subregions of a city are modeled locally while exchanging information through a communication layer—could improve scalability and enable deployment in real-world C-ITS environments with limited centralized infrastructure.

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Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

Ethics Approval Not applicable.

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